Title Page

Title: Formal Lab Report

Sample: A

Name: Robert Taylor

TA: Dominic Fico

Section: 003; L

Introduction

The purpose of the formal lab report is to use spectral data such as NMR, MS< and IR to analyze and identify an unknown compound. Organic chemists use this skill all the time with the use of the real-world skill of reading graphs. The numbers are very physical and can help identify two different compounds. This skill is taking the things we have learned in class and making them reality to see data and be able to apply it and not just memorize or use it and lose it.

Body

Nuclear magnetic resonance (NMR) is one of the many tools used by professional scientist and one of the ones used to identify the unknown. It gives insight to the arrangement and structure of the atoms without skewing the sample given. The sample that is given does not have to weigh very much therefore odd mass numbers can be used. NMR uses specific standards in its testing like the carbon-13 NMR and the proton NMR.

In NMR a magnetic field and radio frequency are applied to the sample to assist in the high spin of the nucleus giving it more energy. The nucleus spins at a very high rate and begins to line up with the magnetic field presented. The nucleus eventually must begin to slow down and lose energy the energy that is lost is then read by the NMR spectra. This is a very complex process using things like decay of atoms to mathematical tools.

Infrared spectroscopy (IR) is another important technique for determining the structure of organic molecules. IR is much more useful not for the overall structure but for understanding what functional groups may be present in a compound. This is done by the separation and importance of single, double, and triple bonds. The mid infrared is the most

important to an organic chemist because this is where typical organic compounds appear. Infrared spectra is donw by takin the sample compound and running it in front of an IR beam. A tool on the other side called a detector sees how much energy is passing through. This can tell how much of the energy is being absorbed by the sample. This energy is mapped on a graph and can identify where certain bonds are on the graph by looking at certain peaks and vibration patterns.

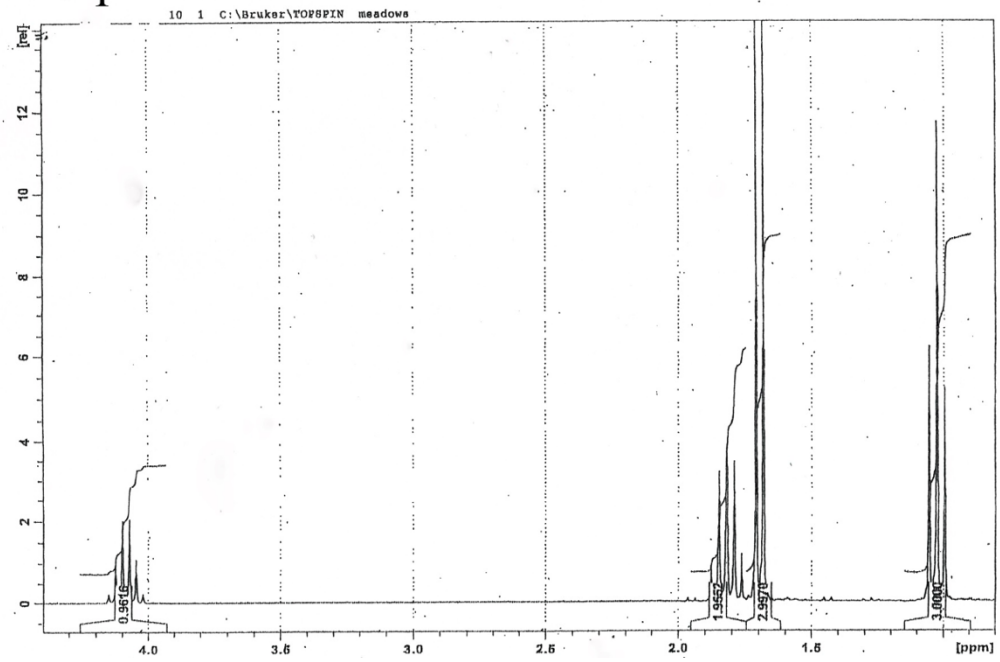
Mass spectroscopy (mass spec.) is the last type of tool used that aides in the identifying of unknown compounds like in lab. The mass spectroscopy does exactly what it sounds like and gives the specific mass of the compound and with this helps to indicate whether certain atoms are in the compound or not. These atoms are of course things like nitrogen, chlorine, and bromine.

Mass spectroscopy takes the compounds and makes them into ions to help see individual characteristics of them. The sample compound is placed in a machine and then flooded with large amounts of high energy electrons. The electrons force movement of certain molecules creating the ions needed to see the specific data. The data gets placed into a set and graphed out by a computer programmed to read the data and then translated onto a line graph. The more intense the peaks the more evident of compounds and the most intense is the base peak.

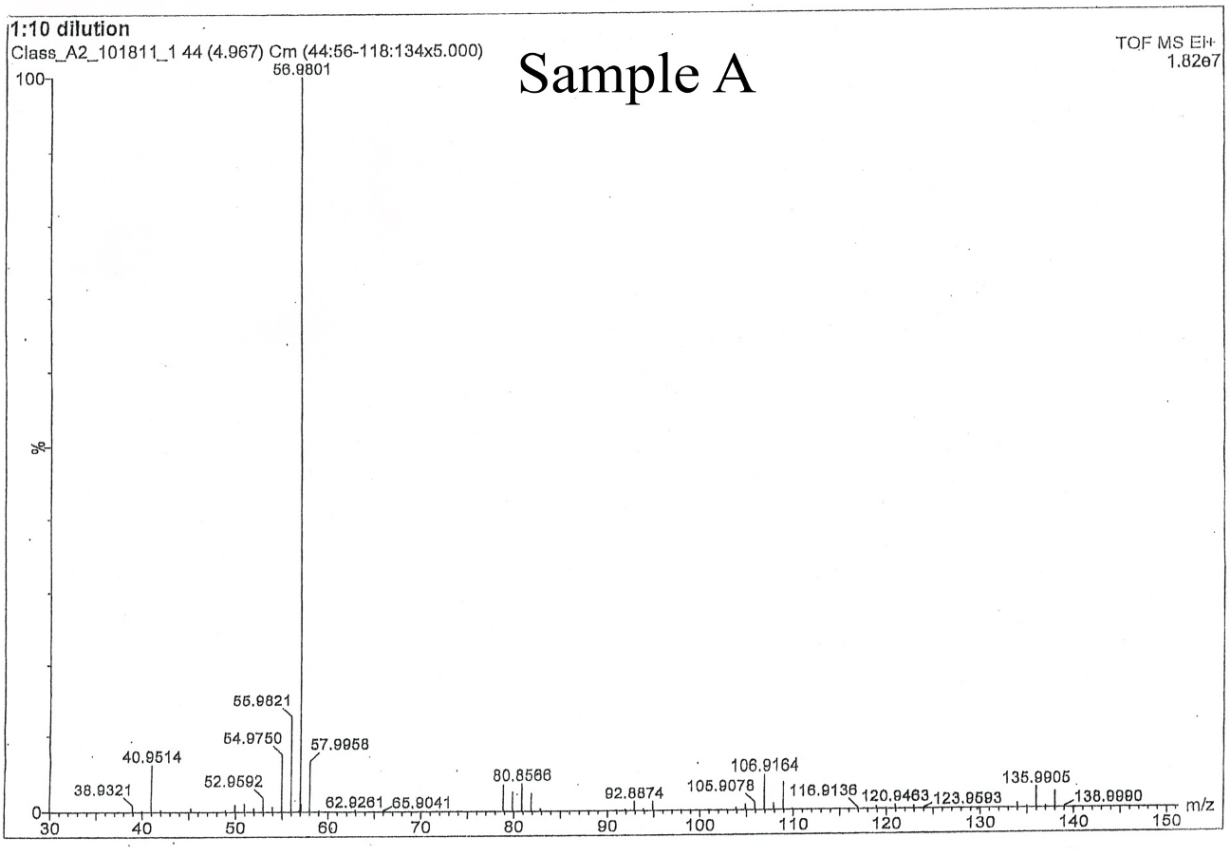
Data

NMR Sample A

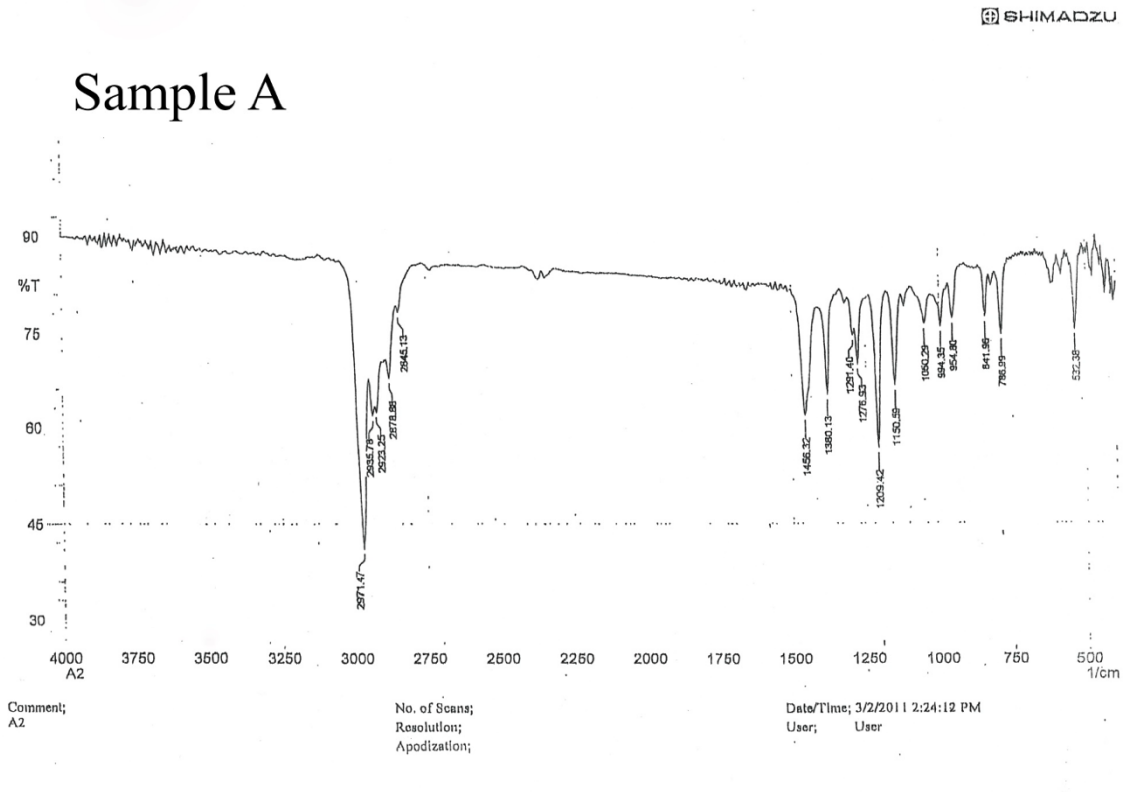
Sample A



GCMS Sample A



FTIR Sample A



Analysis

In the spectra looking specifically at the NMR graph, there are four Individual peaks these peaks are known as four hydrogen atom locations. This helps me to understand that the compound is a basic four carbon structure. There are two doublets seen on the graph. The GCMS indicated a molar mass of 138 grams with the major base peak at around 56.9. This loss shows that the atom loss was a bromine so now it seems the compound is a four-carbon chain with a bromine atom attached. The FTIR spectra shows three major peaks with the largest being at around 2971.74 and is distinctive of an alkane. The third peak seen is between 500-600 range proving a halogen is present and as mentioned earlier is beloved to be a bromine. The compound is identified as a 2-Bromobutane.



References

CHEM 2081: Organic Chemistry II Lab Manual: Ruychev; Department of Chemistry Auburn University; 2023 ed; Hayden-McNeil Publishing

Klein, David. Organic Chemistry; Wiley